Q1: If $22^3 + 23^3 + 24^3 + \ldots$ [illegible] divided by 110, then the remainder will be __ ?

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22, 23, 24 - - - 87, 88 → Arithmetic progression

$$\boxed{a_n = a + (n-1)d}$$

$88 = 22 + (n-1)1$

$88 - 22 = n - 1$

$66 = n - 1$

$n = 66 + 1$

$n = 67$

d: common difference

a: first term

a_n: n^{th} term

n: number of terms

d = बाद वाला term − पहले वाला term

$23 - 22 = 1$

$24 - 23 = 1$

d = [illegible]

Simplification

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Q1: If $22^3 + 23^3 + 24^3 + \ldots\ldots + 87^3 + 88^3$ is divided by 110, then the remainder will be __ ?

$$\boxed{\frac{a^n + b^n}{a+b} \quad n: \text{odd}}$$

$$\frac{a+b+c+d}{110} = \frac{a}{110} + \frac{b}{110} + \frac{c}{110} + \frac{d}{110}$$

$$\frac{a+b}{110} + \frac{c+d}{110}$$

$$\frac{22^3 + 88^3}{22+88} + \frac{23^3 + 87^3}{23+87} + \frac{24^3 + 86^3}{24+86} + \ldots\ldots + \frac{55^3}{110}$$

Remainder : $\frac{55^3}{110} = \frac{55 \times 55 \times 55}{110}$

Answer = 1

Simplification

Q2: If $16^7 + 17^7 + 18^7 + 19^7$ is divided by 5, 7, 14 and 35 and remainder thus obtained are R1, R2, R3 and R4 respectively. Fi

R1 + R2 + R3 + R4 .

$n = 7 \Rightarrow$ odd

$$\frac{a^n + b^n + c^n + d^n}{a + b + c + d}$$

$$\frac{16^7 + 17^7 + 18^7 + 19^7}{16 + 17 + 18 + 19}$$

$$= \frac{16^7 + 17^7 + 18^7 + 19^7}{70} \Rightarrow \frac{\square}{5} \Rightarrow \frac{\square}{7} \Rightarrow \frac{\square}{14}$$

$\therefore R_1 = R_2 = R_3 = R_4 = 0 \Rightarrow R_1 + R_2 + R_3 + R_4$

Simplification 5:59:30 / 11:50:56 TCS Simplification

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$$0.1234 = \frac{1234}{10000}$$

$$0.\overline{1234} = \frac{1234}{9999} \Rightarrow 0.\underbrace{1234}\underbrace{1234}\underbrace{1234}\ldots$$

$$0.1\overline{234} = \frac{1234-1}{9990} \Rightarrow 0.1\underbrace{234}\underbrace{234}\underbrace{234}\ldots$$

$$0.12\overline{34} = \frac{1234-12}{9900} \Rightarrow 0.12\underbrace{34}\underbrace{34}\underbrace{34}\ldots$$

$$0.123\overline{4} = \frac{1234-123}{9000} \Rightarrow 0.123\underbrace{4}\underbrace{4}\underbrace{4}\underbrace{4}\underbrace{4}\ldots$$

Simplification
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Q3: Simplest form $0.\overline{5}$ =

(A) 5/10 (B) 5/9 (C) 1/2 (D) None of these

$0.\overline{5} = \frac{5}{9}$

Simplification

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Q4:

Let $A = 0.a_1a_2a_1a_2a_1a_2....$ and $B = b_1b_2b_1b_2b_1b_2...$ where a_1, a_2, b_1 and b_2 are integers between 1 to 9 not necessarily distinct.

Then with which of the following A must be multiplied such that it gives an integer?

a) 99 b) 198 c) 297 d) All the above

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$A = 0.a_1a_2a_1a_2a_1a_2... = 0.\overline{a_1a_2} = \frac{a_1a_2}{99} = \boxed{\frac{Integer}{99}}$

$99 \times A = \boxed{\frac{Integer}{\cancel{99}}} \times \cancel{99}$

$\boxed{A * 99 = Integer}$

$A * 198 = \frac{Integer}{\cancel{99}} \times \overset{2}{\cancel{198}} = 2 * Integer = Integer$

$A * 297 = \frac{Integer}{\cancel{99}} \times \overset{3}{\cancel{297}} = 3 * Integer = Int$

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Simplification

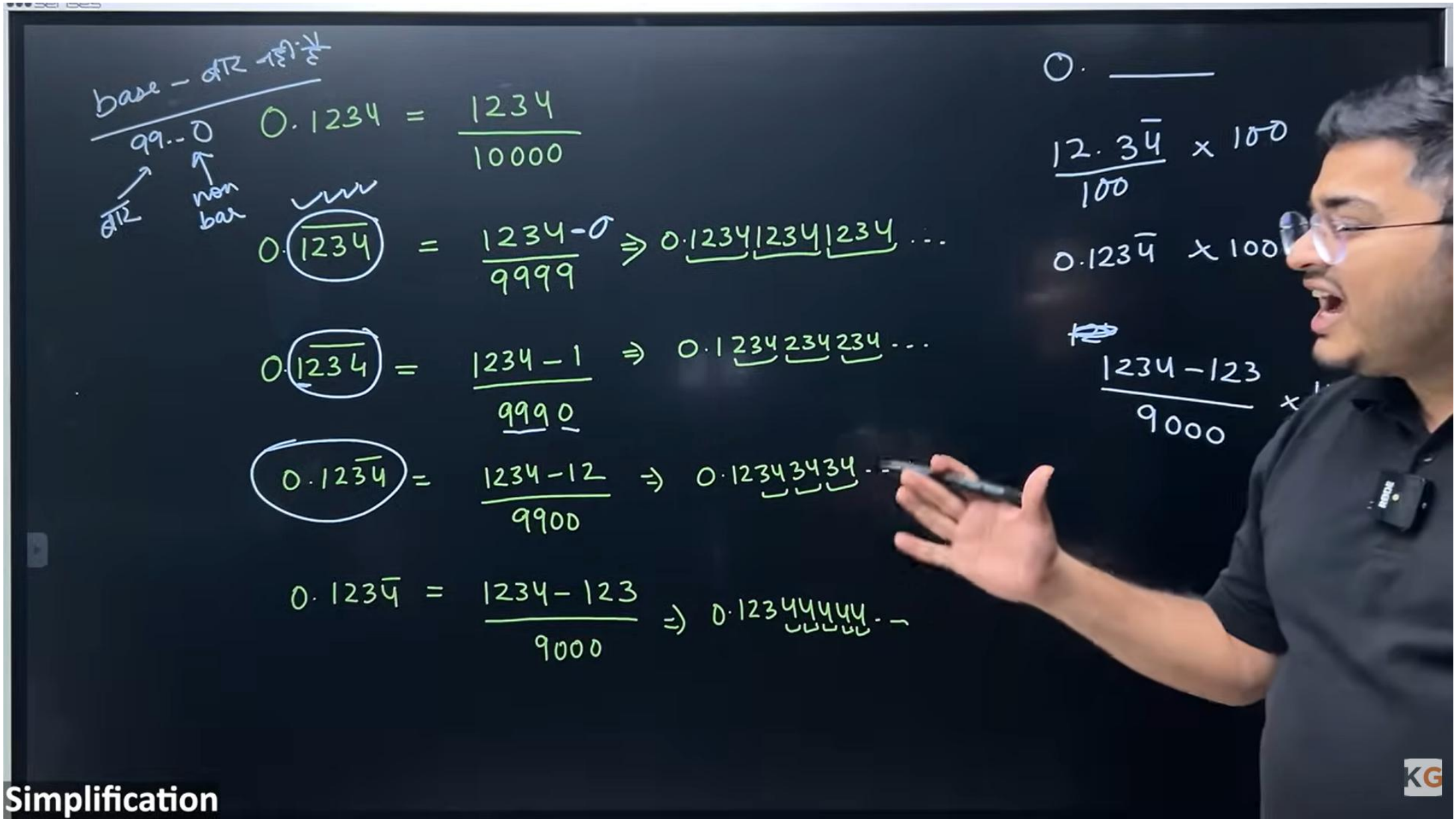
$$A+B = \frac{a_1 a_2 a_3}{999} + \frac{b_1 b_2}{99}$$

$$A+B = \frac{1}{9}\left[\frac{a_1 a_2 a_3}{111} + \frac{b_1 b_2}{11}\right]$$

$$A+B = \frac{1}{9}\left[\frac{11\, a_1 a_2 a_3 + 111\, b_1 b_2}{111 \times 11}\right]$$

$$A+B = \frac{\text{Integer}}{9 \times 111 \times 11} \Rightarrow A+B = \frac{\text{Integer}}{10989} \Rightarrow \boxed{(A+B) \times 10989 = \text{Integer}}$$

Laser Pointer
Pen
Highlighter
Eraser
Erase All Ink on Slide



Simplification